

Optimizing Fruit Yield Prediction: Evaluating Multi-Object Tracking Algorithms for Calamansi Fruit Detection Using YOLOv8m

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Abstract—Accurate fruit yield prediction is necessary for enhancing agricultural methods and providing quality output. This work analyzes the efficacy of multi-object tracking (MOT) algorithms in detecting calamansi fruits from drone-captured video footage, using the YOLOv8m object detection model. Data were gathered from a commercial calamansi farm in Naujan, Philippines, and prepared for model training. Three MOT algorithms—SORT, DeepSORT, and ByteTrack—were evaluated using performance metrics such as Multiple Object Tracking Accuracy (MOTA), Higher Order Tracking Accuracy (HOTA), and ID F1 Score (IDF1). ByteTrack achieved exceptional performance with MOTA, HOTA, and IDF1 ratings of 59.67%, 65.42%, and 73.72% respectively. The integration of YOLOv8m for detection with ByteTrack for tracking significantly reduced double-counting while ensuring good computational efficiency. The results underscore ByteTrack's capability in real-time fruit monitoring and its significance for improving yield estimate and agricultural production.

Keywords—fruit yield prediction, multi-object tracking, YOLOv8m, ByteTrack

I. INTRODUCTION

Sustainable yield in agriculture is crucial, as inappropriate use of agricultural inputs and environmental factors such as weather and soil conditions have adversely affected crop output [1]. Numerous researchers have undertaken efforts to establish accurate agricultural yield assessment methodologies needed for farm expansion and meeting market demands [2][3][4].

Currently, the emergence of advanced cameras and robust processors enables artificial intelligence methodologies, such as deep learning to provide a range of opportunities for real-time object detection [5].

Calamansi, scientifically known as *Citrofortunella microcarpa*, is a citrus fruit cultivated in the Philippines. The production of calamansi is economically vital for regional welfare and the livelihood of local farmers, due to its adaptability to many ecological conditions and its substantial demand in both local and international markets [6]. Autonomous technologies that identify and measure fruits pre-harvest present opportunities for enhancing harvesting procedures [7]. The standard yield measurement methods are manual counting and vision identification. These procedures are, however, subjective,

complicated, and open to human error [8]. Assessing yield accurately is essential for the effective crop management of calamansi producers when evaluating its economic value [9].

Recent technological breakthroughs provide unique instruments that utilize their capabilities for many agricultural purposes. Deep Learning (DL) has demonstrated significant progress in agriculture by executing essential tasks such as yield estimation, crop classification, and fruit detection [6].

Fruit detection provides a specific instance of object detection. In recent years, the majority of methodologies employ end-to-end trained Convolutional Neural Networks (CNNs) for detection purposes [10]. Numerous architectures exist, although the majority of approaches may be categorized into two kinds: single-stage and two-stage models [11]. In recent years, the single stage You Only Look Once (YOLO) family has become a highly favored option in numerous applications owing to its exceptional accuracy, real-time performance, and ease of use [12]. YOLO algorithm has attained an exceptional balance between speed and precision which is needed for accurate detection of objects.

The YOLO series, as a prominent deep learning network, is extensively used for the detection and counting of fruits and vegetables [13]. The average detection accuracy of 92.23% was achieved by YOLOv4 for apple detection [14]. YOLOv5 was applied to autonomously differentiate between fruits that are graspable and those that are ungraspable in apple tree images for a harvesting robot. There are also performance disparities among YOLO models of varying versions. YOLOv5 significantly enhanced the model's mean Average Precision (mAP) by 14.95% and 4.74% compared to YOLOv3 and YOLOv4, respectively, as revealed by comparative analysis [15]. In order to enhance the accuracy of detection, a blueberry recognition method that has been developed from YOLOv8 was applied. The approach achieved a precision of 84.6%, a recall of 91.3%, and a mAP of 94.1% [16]. YOLOv8's competitiveness in mAP@50 scores suggests its potential as a viable option in the object identification domain, despite the fact that YOLOv5 is superior in quantitative measurements. Due to its superior inference times, YOLOv5 is advantageous for real-time applications. Nevertheless, YOLOv8's commendable inference times are also substantial, indicating that it is suitable for

scenarios in which slightly extended processing is reasonable [17].

The YOLOv8 architecture's improvements were also illustrated by other YOLOv8 variants, which outperformed traditional models. Mask R-CNN, FCIS, and Faster R-CNN were conventional models that demonstrated lower efficacy, particularly in the mAP50-95 metric. The models were less robust across a variety of IoU thresholds, as indicated by this. The YOLOv8 variants, in general, exhibited significant improvements over traditional models, highlighting the architecture's adaptability and advancements [18].

Counting fruits poses a challenge, especially because of fruit occlusions [19]. Fruits may appear markedly different due to variations in lighting and storage conditions, complicating the ability of algorithms to reliably identify them in diverse environments. One strategy to address this difficulty entails obtaining data from many views, positing that calamansi fruits obscured from one viewpoint will be observable from another [20]. However, this method may result in an overestimated fruit count due to double counting from various image angles.

Multi-object tracking (MOT) is a computer vision challenge that seeks to recognize and locate objects of interest within a video sequence, associating them across frames to monitor their motions over time [21]. Different trackers adhere to the tracking-by-detection paradigm, which initially localizes and classifies all targets, then associating detections of the same object by appearance, motion, or a combination of both [22][23]. Yield estimates necessitate monitoring methodologies to ensure that objects are counted only once in every frame [24].

An advanced information system for automated fruit detection and counting is important to enhance the productivity of the farm. This study proposes the utilization of multi-view sensing through the acquisition of continuous motion films using a drone in capturing the tree. Thereafter, fruits are identified in every video frame, and Multi-Object Tracking (MOT) methods are employed to avoid double-counting of fruits.

This study compares the SORT, DeepSORT and ByteTrack algorithms for counting fruit in recorded videos. The paper contrasts the three leading tracking methodologies for fruit counting. The following portions of this manuscript are structured as follows: Section II highlights the Materials and Methodology utilized for this research and provides a synopsis of the algorithms applied in the methodological framework. Section III outlines and analyzes the outcomes achieved in fruit detection, tracking, and counting tasks. Finally, in Section IV, the conclusions were defined based on the results of the study.

II. MATERIALS AND METHODOLOGY

A. Dataset

The data was gathered from calamansi farms in Naujan, Oriental Mindoro, particularly at Calangi Farm in Aurora, Naujan. The farm engages in commercial production aimed at broader markets, including exports. It is 5 hectares and has more than 3000 calamansi trees. The researchers gathered data from 300 trees on the land, which were shot from January to February 2024 between 8 AM-11 AM, and 2 PM-4 PM. The fruit trees and flowers were photographed from many angles and under

varying lighting conditions to collect sample data for research purposes.

RGB video frames were recorded with a DJI Air 25 drone mounted with a 20 MP camera, attaining a video resolution of 5472×3078 at 25fps in calamansi fields during daylight hours. The video clips were stored on an SD card and transferred to the local computer for image segmentation. The video frames were converted into pictures via Roboflow. Fig. 1 illustrates the topographic and lateral perspectives of the tree acquired by the drone.

The drone maintained a distance of 1 to 1.5 meters from the target, with the camera positioned vertically upward or below while either hovering or gradually navigating around the tree. This dynamic movement reveals the tree's connection with its surroundings. Considering the typical height of Calamansi trees, which ranges from 3 to 6 meters, a minimum altitude of 4 to 8 meters over the tallest tree in the orchard was maintained to guarantee safe flying and avert any harm to both the drone and the trees. This offered detailed views of the Calamansi trees, including close-ups of the fruits and flowers.



Fig. 1. Example data captured by the drone

B. Object Detection Model

This research employs the YOLOv8m object identification convolutional neural network to detect fruits in each frame of the video clip. The YOLOv8m model was trained using customized datasets. Two classes were used in the training: flowers and fruits. The data was subjected to pre-processing and annotation using the web-based image annotation tool, Roboflow. Table 1 shows the parameters used in model training.

TABLE I. TRAINING PARAMETERS

Parameters	Setup
Epochs	100
batch	32
Imagesz	640
Optimizer	SGD
Initial Learning Rate	0.01
Patience	20

The dataset was assessed with three different iterations of YOLOv8 to identify the most suitable version for model training. The dataset was used for training and testing the YOLOv8n, YOLOv8s, and YOLOv8m models. An analysis of three iterations of the YOLOv8 model—YOLOv8n, YOLOv8s, and YOLOv8m—indicated that the YOLOv8m model consistently exhibits enhanced performance across many metrics. The model demonstrated a precision score of 0.839,

indicating its ability to provide positive predictions with significant accuracy. It was also compared with Roboflow 3.0 as seen in Table 2.

TABLE II. EVALUATION RESULTS OF YOLOV8M

Model Version	Precision	Recall	F1 score	mAP@0.5
Roboflow 3.0	0.774	0.668	0.717	0.724
YOLOV8m	0.839	0.794	0.815	0.875

A comparative analysis of three MOT trackers—SORT, DeepSORT, and ByteTrack—for the fruit tracking was conducted. SORT depends significantly on precise object detections for data association. It employs many association techniques, including the Kalman Filter and Hungarian Algorithm, which depend on a motion model of identified objects. DeepSORT enhances SORT by integrating appearance data derived from a re-identification (Re-ID) network. ByteTrack uses both high- and low-confidence detections to proficiently manage densely clustered objects or complex situations.

C. Object Counting

The YOLOv8m model detects objects in each video frame and provides the bounding box coordinate and class ID to indicate the type of object (flower or fruit). After extracting the bounding box coordinates for each detection, each detected object is passed to the tracker. The tracker assigns a unique ID to each object that it detects and tracks across frames. This process helps maintain the ID of each object across frames, even if it moves.

Each detected object now has an assigned unique ID provided by the tracker. For each tracked object in the frame, the system extracts the object's unique ID and class label. It checks if the object's ID has already been counted for that class. If the ID is new (not yet counted), the count of the object will be added. Each unique object is counted only once for each class across all frames. Fig. 2 shows the example tracked objects and their unique IDs.



Fig. 2. Example of tracked objects

D. Evaluation Metrics

To evaluate the performance of the tracking algorithms, a set of standard metrics was used in multi-object tracking (MOT) tasks. These metrics help assess how well the tracker follows objects over time, handles object re-identification, and manages various tracking challenges like occlusions.

To measure the overall accuracy of the tracking system by considering the effects of false positives, missed detections, and

mismatches, the study used Multiple Object Tracking Accuracy (MOTA).

To identify the harmonic mean of precision and recall, based on unique identities rather than bounding boxes, the ID F1 Score (IDF1) was used.

Higher Order Tracking Accuracy (HOTA) is an expansion of the MOTA that integrates many metrics to assess numerous error kinds, including detection recall and accuracy, association recall and precision, and localization.

These metrics (MOTA, HOTA, IDF1) evaluate the tracker's ability to maintain consistent IDs, detect and follow objects accurately, and process frames efficiently.

III. RESULTS AND DISCUSSIONS

Table 2 presents a comparative analysis of the findings obtained by SORT, DeepSORT and ByteTrack on the modified dataset. ByteTrack achieved superior performance metrics, achieving MOTA, HOTA and IDF1 of 59.671%, 65.424%, and 73.72%, respectively. ByteTrack outperformed the other two trackers, demonstrating substantial performance improvement. By considering the aforementioned data, it was determined that ByteTrack achieves consistent behavioral monitoring of fruits and flowers.

TABLE III. TRAINING PARAMETERS

Tracker	MOTA	HOTA	ID F1
SORT	41.59	31.42	44.52
DeepSORT	51.57	51.20	61.961
ByteTrack	59.671	65.424	73.72

According to these data, ByteTrack is the most effective approach among the three trackers. It attains superior accuracy metrics (MOTA, IDF1, and HOTA) relative to SORT and DeepSORT, while preserving comparable computational efficiency to SORT. Consequently, ByteTrack provides an advantageous blend of exceptional tracking efficacy and moderate computing expense, making it the preferred option for fruit monitoring applications.

Furthermore, end users must evaluate the necessity of providing the required hardware in implementing the system. The costs will vary based on the hardware models that will be used as shown in Table IV.

TABLE IV. COST ANALYSIS

Category	Low Estimate	High Estimate
Drone hardware (DJI Mini 2 SE- DJI Phantom 4 RTK)	18, 590.00	182, 064.00
Laptop hardware (Acer Aspire 6- MSI Stealth 15M)	38, 949.00	64, 950.00

A fruit counting system using computer vision, specifically leveraging YOLOv8m and ByteTrack tracking techniques, shows efficacy in calculating farm productivity. The integration of sophisticated data analysis and deep learning methodologies

provided an opportunity to create models for forecasting fruit harvests.

By employing a Structured Query Language (SQL) database to implement this system in the future, it would be advantageous to manage extensive datasets. Organizations can guarantee efficient and reliable data processing by implementing strategies such as indexing, partitioning, sharding, and cache, which considerably improve their database performance [25]. The model could only detect calamansi fruits and flowers because the data used to train the model were customized. The detected objects will be stored in the database. By means of partitioning, replication, and cloud services, SQL is capable of accommodating the increasing demands of data. This would be particularly advantageous when managing larger datasets.

The development of a fruit yield prediction system for producers has the potential to enhance agricultural management, increase revenue, and reduce risk and losses. By predicting produce volumes in advance, farmers can more efficiently allocate resources such as labor, water, fertilizers, and pesticides, thereby reducing waste and superfluous expenditures. The system assists the farmer in the more effective organization of harvest and post-harvest activities, such as storage, transport, and market delivery. In conclusion, a fruit yield prediction system enables farmers to make informed decisions, maximize their operations, and mitigate risks, thereby improving their economic stability and profitability.

IV. CONCLUSION

This study used computer vision to develop and evaluate a fruit counting system intended to predict orchard productivity. The technology was assessed in an experimental calamansi orchard. The fruit count in recorded videos was primarily compared to the efficacy of three advanced tracking methods: SORT, DeepSORT, and ByteTrack. Across performance metrics, ByteTrack consistently excels achieving superior results in MOTA, IDF1, and HOTA relative to DeeSORT, however SORT had less effective performance. Nonetheless, these findings suggest that the appearance characteristics of the SORT algorithm do not significantly assist fruit tracking.

This research is a vital phase in the process of converting theoretical developments in MOT and deep learning into practical agricultural solutions. By utilizing drone-based data collection and processing, the research illustrates the potential of cutting-edge technology to predict fruit yields in a scalable and efficient manner, thereby bridging the divide between computational theory and real-world applications.

This research is a pioneer in the incorporation of sophisticated tracking paradigms that are specifically designed for agricultural environments by utilizing MOT algorithms—SORT, DeepSORT, and ByteTrack. The research demonstrates that the integration of sophisticated tracking mechanisms with high-precision object detection can result in more accurate predictions of fruit yield, even in visually complex and dynamic environments such as orchards.

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REFERENCES

- [1] Vogel, E., Donat, M. G., Alexander, L. V., Meinshausen, M., Ray, D. K., Karoly, D., Meinshausen, N., & Frieler, K. (2019). The effects of climate extremes on global agricultural yields. *Environmental Research Letters*, 14(5), 054010. <https://doi.org/10.1088/1748-9326/ab154b>
- [2] Hu, N., Su, D., Wang, S., Nyamsuren, P., Qiao, Y., Jiang, Y., & Cai, Y. (2022). LettuceTrack: Detection and tracking of lettuce for robotic precision spray in agriculture. *Frontiers in Plant Science*, 13. <https://doi.org/10.3389/fpls.2022.1003243>
- [3] Saddik, A., Latif, R., Abualkashik, A. Z., Ouardi, A. E., & Elhoseny, M. (2023b). Sustainable yield prediction in agricultural areas based on fruit counting approach. *Sustainability*, 15(3), 2707. <https://doi.org/10.3390/su15032707>
- [4] Jhahharia, K., Mathur, P., Jain, S., & Nijhawan, S. (2023). Crop Yield Prediction using Machine Learning and Deep Learning Techniques. *Procedia Computer Science*, 218, 406–417. <https://doi.org/10.1016/j.procs.2023.01.023>
- [5] Butera, L., Ferrante, A., Jermini, M., Prevostini, M., & Alippi, C. (2021). Precise Agriculture: Effective deep learning strategies to detect pest insects. *IEEE/CAA Journal of Automatica Sinica*, 9(2), 246–258. <https://doi.org/10.1109/jas.2021.1004317>
- [6] Quijano, M. F., Quijano, G., & Diaz, R. (2021). Agricultural economic production of Philippine Calamansi industry: A basis for production local development plan. <https://doi.org/10.20944/preprints202102.0388.v1>
- [7] Mureşan, H., Călin, A. D., & Coroiu, A. M. (2020). Overview of recent deep learning methods applied in fruit counting for yield estimation. *Studia Universitatis Babes-Bolyai: Series Informatica*, 65(2), 50. <https://doi.org/10.24193/subbi.2020.2.04>
- [8] De Luna, R. G., Magnaye, V. C., Reaño, R. a. L., Enriquez, K. T., Armando, R. R., Bocalbos, M. L., Fernández, J. C. F., Malacaman, K. A., Natividad, J., Ramos, J., & Salcedo, S. M. (2023). A Machine Learning-Based Approach for Accurate Size Classification of Pineapple (Ananas Comosus). *TENCON 2023 - 2023 IEEE Region 10 Conference (TENCON)*. <https://doi.org/10.1109/tencon58879.2023.10322473>
- [9] Vignesh, K., Askarunisa, A., & Abirami, A. M. (2023). Optimized deep learning methods for crop yield prediction. *Computer Systems Science and Engineering*, 44(2), 1051–1067. <https://doi.org/10.32604/csse.2023.024475>
- [10] Dhillon, A., & Verma, G. K. (2019). Convolutional Neural Network: A review of models, methodologies and applications to object detection. *Progress in Artificial Intelligence*, 9(2), 85–112. <https://doi.org/10.1007/s13748-019-00203-0>
- [11] Du, L., Zhang, R., & Wang, X. (2020d). Overview of two-stage object detection algorithms. *Journal of Physics Conference Series*, 1544(1), 012033. <https://doi.org/10.1088/1742-6596/1544/1/012033>
- [12] Diwan, T., Anirudh, G., & Tembhurne, J. V. (2022). Object detection using YOLO: challenges, architectural successors, datasets and applications. *Multimedia Tools and Applications*, 82(6), 9243–9275. <https://doi.org/10.1007/s11042-022-13644-y>
- [13] Jiang, P., Ergu, D., Liu, F., Cai, Y., & Ma, B. (2022b). A review of Yolo algorithm developments. *Procedia Computer Science*, 199, 1066–1073. <https://doi.org/10.1016/j.procs.2022.01.135>
- [14] Wang, Z., Jin, L., Wang, S., & Xu, H. (2021b). Apple stem/calyx real-time recognition using YOLO-v5 algorithm for fruit automatic loading system. *Postharvest Biology and Technology*, 185, 111808. <https://doi.org/10.1016/j.postharvbio.2021.111808>

- [15] Yan, B., Fan, P., Lei, X., Liu, Z., & Yang, F. (2021c). A Real-Time Apple targets detection method for picking robot based on improved YOLOV5. *Remote Sensing*, 13(9), 1619. <https://doi.org/10.3390/rs13091619>
- [16] Gai, R., Liu, Y., & Xu, G. (2024b). TL-YOLOV8: A Blueberry fruit detection algorithm based on improved YOLOV8 and transfer learning. *IEEE Access*, 12, 86378–86390. <https://doi.org/10.1109/access.2024.3416332>
- [17] Casas, E., Ramos, L., Bendek, E., & Rivas-Echeverria, F. (2024). YOLOv5 vs. YOLOv8: Performance Benchmarking in Wildfire and Smoke Detection Scenarios. *Journal of Image and Graphics*, 12(2), 127–136. <https://doi.org/10.18178/joig.12.2.127-136>
- [18] Maulana, A. W., Wibowo, S. A., Sunarya, U., Rahmania, R., & Insani, A. (2024). Performance improvement of YOLOV8-RTDETR method based retail product detection. *Journal of Image and Graphics*, 12(3), 292–301. <https://doi.org/10.18178/joig.12.3.292-301>
- [19] Gené-Mola, J., Ferrer-Ferrer, M., Gregorio, E., Blok, P. M., Hemming, J., Morros, J., Rosell-Polo, J. R., Vilaplana, V., & Ruiz-Hidalgo, J. (2023b). Looking behind occlusions: A study on amodal segmentation for robust on-tree apple fruit size estimation. *Computers and Electronics in Agriculture*, 209, 107854. <https://doi.org/10.1016/j.compag.2023.107854>
- [20] Rapado-Rincón, D., Van Henten, E. J., & Kootstra, G. (2023). Development and evaluation of automated localisation and reconstruction of all fruits on tomato plants in a greenhouse based on multi-view perception and 3D multi-object tracking. *Biosystems Engineering*, 231, 78–91. <https://doi.org/10.1016/j.biosystemseng.2023.06.003>
- [21] Ariza-Sentís, M., Vélez, S., Martínez-Peña, R., Baja, H., & Valente, J. (2024b). Object detection and tracking in Precision Farming: a systematic review. *Computers and Electronics in Agriculture*, 219, 108757. <https://doi.org/10.1016/j.compag.2024.108757>
- [22] Yang, H., Chang, F., Huang, Y., Xu, M., Zhao, Y., Ma, L., & Su, H. (2022). Multi-object tracking using Deep SORT and modified CenterNet in cotton seedling counting. *Computers and Electronics in Agriculture*, 202, 107339. <https://doi.org/10.1016/j.compag.2022.107339>
- [23] Duong-Trung, H., & Duong-Trung, N. (2024). Integrating YOLOV8-Agri and DeepSORT for advanced motion detection in agriculture and fisheries. *EAI Endorsed Transactions on Industrial Networks and Intelligent Systems*, 11(1). <https://doi.org/10.4108/eetinis.v11i1.4618>
- [24] Ciarfuglia, T. A., Motoi, I. M., Saraceni, L., Fawakherji, M., Sanfeliu, A., & Nardi, D. (2023). Weakly and semi-supervised detection, segmentation and tracking of table grapes with limited and noisy data. *Computers and Electronics in Agriculture*, 205, 107624. <https://doi.org/10.1016/j.compag.2023.107624>
- [25] Uzzaman, A., Jim, M. M. I., Nishat, N., & Nahar, J. (2024). OPTIMIZING SQL DATABASES FORBIG DATA WORKLOADS: TECHNIQUES AND BEST PRACTICES. *Academic Journal on Business Administration, Innovation & Sustainability*, 4(3), 15–29. <https://doi.org/10.69593/ajbais.v4i3.78>